

Isaac Mubiru

isaacmubiru99@gmail.com | linkedin.com/in/isaac-mubiru-3bb728174 | github.com/zac-09 | Kampala, Uganda (UTC+3) — Fully Remote

PROFESSIONAL SUMMARY

Senior fullstack engineer with 5+ years building scalable consumer-based apps in Node.js/TypeScript and React. Led the zero-downtime migration of 20+ production applications from Parse/MongoDB to Firebase on Google Cloud, including a real-time two-way MongoDB sync pipeline over Pub/Sub — cutting \$5,000/month in infrastructure costs. Equally at home in the browser and the Node.js runtime: shipped responsive React/Tailwind PWAs serving real-time stock prices from 2M+ database records, consumer-facing media features built on ffmpeg, and REST APIs on Google Cloud. Fully distributed-team native — 5 years of async remote delivery across US and Singapore teams.

CORE COMPETENCIES

Node.js & TypeScript

React / HTML / CSS

MongoDB & SQL

Google Cloud (GCP)

REST API Design

Consumer-Scale Performance

Integration Testing & Zero-Downtime Releases

Kubernetes & Docker

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer — US Remote (consumer e-commerce)

- Led the end-to-end migration of **20+ consumer applications** from Parse/MongoDB to Firebase/Google Cloud with **zero downtime**, reporting directly to the CTO — staged rollouts validated with record-level integration diffs before each cutover
- Built a **real-time two-way sync between MongoDB and Firestore** using Node.js and Google Pub/Sub, keeping live consumer traffic consistent across both databases for the duration of the migration
- Shipped a consumer-facing **3D visualisation feature** using video overlay and ffmpeg, increasing buyer conversion
- Implemented Express Shipping, generating an additional \$40 revenue per order
- Saved the company **\$5,000/month** by migrating all services off AWS; moved file storage from S3 to Google Cloud Storage
- Migrated a Webflow website from a Parse backend to Firebase

MTailor (Contract)

Mar 2022 – Jul 2022

Backend Contractor — US Remote

- Built the MongoDB-to-Firestore migration pipeline with complex in-flight processing logic; promoted to full-time on delivery
- Wrote engineering documentation for the new Firebase SDKs (Node.js and Python) and trained the Ops team on dashboards and backups

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer (Fullstack) — Singapore Remote (consumer fintech)

- Led optimisation of a consumer **PWA delivering real-time stock market prices** from a Firebase backend
- Built responsive UI pages with **React, Tailwind, and HTML/CSS** for the production PWA
- Built ad-hoc jobs querying **2M+ Firestore records** to keep customer prices current from Morningstar APIs
- Extended backend **REST APIs** using Firebase Cloud Functions and Express.js hosted on Heroku

Mind2matter.co

Jan 2022 – Jul 2022

Software Engineer (Fullstack) — US Remote

- Delivered pixel-accurate **React UIs from Figma designs** under tight agency timelines
- Built backends for DeFi applications using Web3 and Node.js

CodeBits

Jan 2020 – Jul 2021

Software Engineer / Team Lead — Uganda Remote (legal tech)

- Led a team of 4 developers building case management and paralegal database systems for NGOs
- Architected a **microservices backend** using Apache Kafka, Docker, and **Kubernetes**
- Built a USSD service communicating with a Node.js backend via gRPC; cross-platform React Native app with Firebase push notifications

PROJECTS

Real-Time Pub/Sub Sync Pipeline

Live MongoDB ↔ Firestore sync supporting parallel production traffic during a months-long migration — idempotent event consumers, no data loss across 20+ apps.

Node.js · Google Pub/Sub · MongoDB · Firestore

Marketplace Recommendation Engine

Collaborative + content-based filtering engine in MongoDB for matching consumers to service providers.

Node.js · MongoDB · aggregation pipelines

Healthcare WhatsApp Chatbot

Prescription OCR and conversational flows over the WhatsApp Business API, integrating OpenAI and Google Vision APIs.

Node.js · REST integrations · Google Cloud Vision

EDUCATION

BSc Computer Engineering — [Makerere University](#)

2018 – 2022

Kampala, Uganda

SKILLS

Proficient: Node.js, TypeScript/JavaScript, React, MongoDB, SQL, Google Cloud (Firestore, Pub/Sub, Cloud Functions, GCS), Firebase, HTML, CSS, Python, Git, Linux

Intermediate: Go, AWS, Rust, React Native, Kubernetes, Docker, Kafka **Languages:** English, Kiswahili